

Problem 15.1.16

Is the series convergent or divergent? Give a reason.

$$\sum_{n=0}^{\infty} \frac{(20 + 30i)^n}{n!}$$

Problem 15.1.17

Is the series convergent or divergent? Give a reason.

$$\sum_{n=2}^{\infty} \frac{(-i)^n}{\ln n}$$

Problem 15.1.18

Is the series convergent or divergent? Give a reason.

$$\sum_{n=1}^{\infty} n^2 \left(\frac{i}{4} \right)^n$$

Problem 15.1.19

Is the series convergent or divergent? Give a reason.

$$\sum_{n=0}^{\infty} \frac{i^n}{n^2 - i}$$

Problem 15.1.20

Is the series convergent or divergent? Give a reason.

$$\sum_{n=0}^{\infty} \frac{n + i}{3n^2 + 2i}$$

Problem 15.1.21

Is the series convergent or divergent? Give a reason.

$$\sum_{n=0}^{\infty} \frac{(\pi + \pi i)^{2n+1}}{(2n+1)!}$$

Problem 15.1.22

Is the series convergent or divergent? Give a reason.

$$\sum_{n=1}^{\infty} \frac{1}{\sqrt{n}}$$

Problem 15.1.23

Is the series convergent or divergent? Give a reason.

$$\sum_{n=0}^{\infty} \frac{(-1)^n (1+i)^{2n}}{(2n)!}$$

Problem 15.1.24

Is the series convergent or divergent? Give a reason.

$$\sum_{n=1}^{\infty} \frac{(3i)^n n!}{n^n}$$

Problem 15.2.7

Find the center and the radius of convergence.

$$\sum_{n=0}^{\infty} \frac{(-1)^n}{(2n)!} \left(z - \frac{1}{2}\right)^{2n}$$

Problem 15.2.9

Find the center and the radius of convergence.

$$\sum_{n=0}^{\infty} \frac{n(n-1)}{3^n} (z-i)^{2n}$$

Problem 15.2.13

Find the center and the radius of convergence.

$$\sum_{n=0}^{\infty} 16^n (z+i)^{4n}$$

Problem 15.2.15

Find the center and the radius of convergence.

$$\sum_{n=0}^{\infty} \frac{(2n)!}{4^n (n!)^2} (z-2i)^n$$

Problem 15.2.17

Find the center and the radius of convergence.

$$\sum_{n=1}^{\infty} \frac{2^n}{n(n+1)} z^{2n+1}$$

Problem 15.3.5

Find the radius of convergence in two ways:

- (a) Directly by the Cauchy–Hadamard formula in Sec. 15.2
- (b) From a series of simpler terms by using Theorem 3 or Theorem 4

$$\sum_{n=2}^{\infty} \frac{n(n-1)}{2^n} (z-2i)^n$$

Problem 15.3.7

Find the radius of convergence in two ways:

- (a) Directly by the Cauchy–Hadamard formula in Sec. 15.2
- (b) From a series of simpler terms by using Theorem 3 or Theorem 4

$$\sum_{n=1}^{\infty} \frac{n}{3^n} (z+2i)^{2n}$$

Problem 15.3.9

Find the radius of convergence in two ways:

- (a) Directly by the Cauchy–Hadamard formula in Sec. 15.2
- (b) From a series of simpler terms by using Theorem 3 or Theorem 4

$$\sum_{n=1}^{\infty} \frac{(-2)^n}{n(n+1)(n+2)} z^{2n}$$

Problem 15.3.11

Find the radius of convergence in two ways:

- (a) Directly by the Cauchy–Hadamard formula in Sec. 15.2
- (b) From a series of simpler terms by using Theorem 3 or Theorem 4

$$\sum_{n=1}^{\infty} \frac{3^n n(n+1)}{7^n} (z+2)^{2n}$$

Problem 15.3.13

Find the radius of convergence in two ways:

- (a) Directly by the Cauchy–Hadamard formula in Sec. 15.2
- (b) From a series of simpler terms by using Theorem 3 or Theorem 4

$$\sum_{n=0}^{\infty} \left[\binom{n+k}{k} \right]^{-1} z^{n+k}$$